Spectral Early Warning Signals for Model Collapse and Their Prevention–Detectability Tradeoff

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Abstract

During the early stages of recursive synthetic training, spectral indicators of the embedding space suggest the model is improving, while downstream performance degrades. We characterize this disconnect by tracking spectral properties across 10 recursive generations of Pythia-410M. The result is a reproducible three-phase pattern consistent across three seeds: spectral decompression (generations 0–2), where spectral log-volume (SLV) increases while HellaSwag drops 3.3 percentage points; collapse onset (generations 2–4), where spectral indicators reverse; and accelerating collapse (generations 4–10). Of eight candidate indicators, SLV is the most effective indicator: it requires one forward pass (< 1 s), captures changes across all spectral directions, and declines monotonically after the decompression peak. Varying the mixing ratio across five levels yields a trajectory map localizing the critical mixing ratio $r_{\text{crit}} \in (0.25, 0.3]$, separating monotonic expansion from a transitional regime with initial contraction then recovery. A proportional controller reveals a prevention–detectability tradeoff: real-data mixing suppresses spectral contraction, so the rate-based signal vanishes precisely when prevention succeeds—a particularly strong form of the ecological intervention–warning tradeoff. Setting $r > r_{\text{crit}}$ prevents collapse without per-generation monitoring. The three-phase trajectory reproduces on GPT-2-medium (355M), providing preliminary cross-architecture evidence[1].

1 Introduction

Synthetic data has become integral to language model training. Instruction tuning, knowledge distillation, and data augmentation all routinely use model-generated text, and as synthetic corpora are recycled into training pipelines, the process often becomes recursive: each generation of models is fine-tuned on outputs from the previous one. Recent work demonstrates that such recursive synthetic training triggers *model collapse*, a progressive erosion of distributional tails and downstream capability (Shumailov et al., 2024; Dohmatob et al., 2024b,a; Alemohammad et al., 2024). The problem is especially pressing as web-scraped corpora increasingly contain model-generated content (Dohmatob et al., 2025), a trend that introduces implicit recursive loops even without deliberate data reuse.

While model collapse is now well-characterized through theoretical analysis (Dohmatob et al., 2024b; Seddik et al., 2024) and empirical studies (Shumailov et al., 2024; Alemohammad et al., 2024; Gerstgrasser et al., 2024), its effect on the geometric structure of learned representations (Ethayarajh, 2019; Martin and Mahoney, 2021) remains poorly understood. Characterizing how the embedding space contracts during recursive training could yield computationally cheap collapse indicators from internal model states alone, without text generation or benchmark evaluation.

Current collapse detection depends on downstream metrics such as perplexity and benchmark accuracy, or on analysis of generated text through lexical diversity measures. Both approaches require either costly inference or external evaluation infrastructure at every generation. SIGMA (Gu et al., 2026) introduced spectral monitoring through embedding covariance analysis and showed that spectral indicators can track collapse, but validated only on OPT-125M over short generation horizons. Four gaps remain open. First, although spectral collapse has been linked to plasticity loss (He et al., 2025) and rank collapse observed in attention layers (Nait Saada et al., 2025), no work has characterized spectral dynamics in recursive training beyond a few generations. Second, the temporal relationship between spectral indicators and downstream degradation has not been examined.

[1]Code and data are available at https://anonymous.4open.science/r/spectral_collapse_ews-1B2C/.

Third, ecological early warning signals (EWS) theory warns that stabilization interventions can suppress the precursors they rely on (Scheffer et al., 2009, 2012); whether this prevention–detectability tradeoff manifests in recursive model training—and how severely—has not been tested. Fourth, while data mixing strategies are well-studied for pretraining (Longpre et al., 2024; Xie et al., 2023), the relationship between real-data mixing ratios and spectral collapse has not been mapped: it is unknown how much real data suffices to prevent spectral contraction, or whether a sharp phase boundary exists.

We address these gaps by tracking spectral indicators of the embedding matrix across 10 recursive generations of Pythia-410M-deduped (Biderman et al., 2023), with pilot validation on Pythia-1B-deduped and cross-architecture confirmation on GPT-2-medium (Radford et al., 2019). Our analysis reveals a reproducible three-phase pattern (Figure 1): spectral decompression (Gen 0–2), where effective rank (r_{eff}) (Roy and Vetterli, 2007) and spectral log-volume (SLV) rise while downstream performance already declines; collapse onset (Gen 2–4), where spectral indicators reverse; and accelerating collapse (Gen 4–10), where contraction accelerates. Among these indicators, SLV is the most effective collapse indicator: it requires only a forward pass on held-out data (no text generation), captures distributed volume changes across all spectral directions, and declines monotonically after its peak. We further design a proportional controller as a testbed for reactive collapse prevention, uncovering a prevention–detectability tradeoff: the rate-based spectral signal degrades continuously as the real-data fraction increases and vanishes entirely above r_{crit}, leaving no regime where reactive control and effective prevention coexist.

Our contributions are:

- **C1:** The first multi-seed spectral-downstream characterization over 10 recursive generations, which reveals dynamics invisible in prior short-horizon studies and establishes a reproducible three-phase collapse pattern (decompression, onset, acceleration) (§4.2, Table 2).
- **C2:** SLV as a computationally cheap collapse indicator with monotonic post-peak decline and a HellaSwag-spectral asymmetry: downstream degradation begins during the spectral decompression phase, so spectral expansion does not indicate model health (§4.2, §4.4).

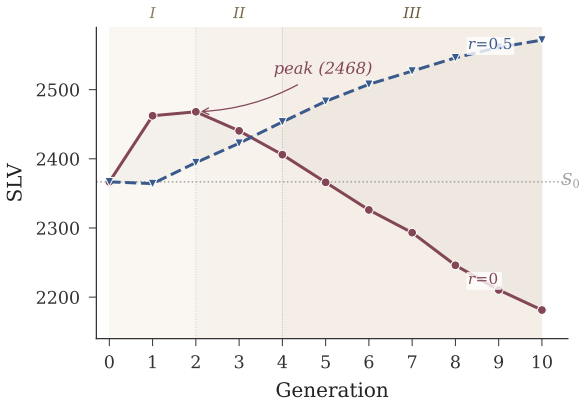


Figure 1: SLV across 10 recursive generations of Pythia-410M (solid: r=0, 3-seed mean; dashed: r=0.5, 2-seed mean) under pure-synthetic training and 50% real-data mixing. Three phases are visible in the pure-synthetic trajectory: decompression (Gen 0–2, SLV peaks at 2468), onset (Gen 2–4), and acceleration (Gen 4–10, SLV falls 11.6% from peak to 2181). Under 50% mixing, SLV rises monotonically (except a negligible Gen 1 dip) to 2572 at Gen 10, 8.7% above baseline and 104 points above the pure-collapse peak.

- **C3:** A prevention–detectability tradeoff: the spectral controller signal degrades continuously as the real-data fraction approaches r_{crit} and vanishes entirely above it—a particularly strong form of the ecological intervention–warning tradeoff and, to our knowledge, the first quantification of this tradeoff in ML (§3.2, §4.3).

Supporting these findings, we construct a spectral collapse trajectory map across multiple mixing ratios that localizes r_{crit} empirically, with the bound robust to both the exponential moving average (EMA) smoothing parameter and random seed (§3.3, §4.3, §4.5), and provide preliminary cross-architecture evidence that the three-phase trajectory generalizes to GPT-2-medium (§4.6).

2 Related Work

Model Collapse. Shumailov et al. (2024) showed that recursive training on model-generated data progressively erodes distributional tails (*model collapse*). Scaling laws predict tail erosion as a function of recursive depth (Dohmatob et al., 2024b,a; Seddik et al., 2024); the self-consuming loop accelerates collapse even under data accumulation (Alemohammad et al., 2024), with as little as 1% synthetic contamination sufficing (Dohmatob et al., 2025), though preserving all prior data can delay it (Gerstgrasser et al., 2024); mitigations include

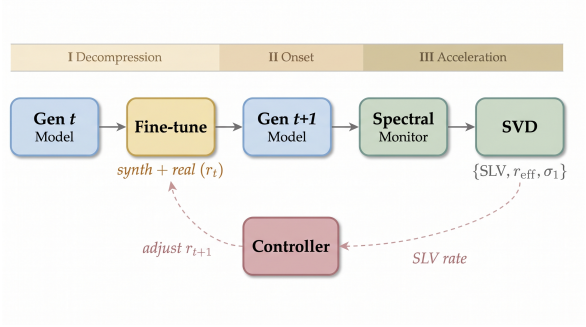


Figure 2: Experimental pipeline. Each generation fine-tunes on synthetic text mixed with real data at ratio r_t. Spectral monitoring extracts last-layer embeddings from held-out data, computes a singular value decomposition (SVD), and derives SLV, r_{eff}, and σ_1. A proportional controller (dashed) can adjust r_{t+1} based on the SLV rate of change. The three phases (decompression (I), onset (II), and acceleration (III)) are annotated along the generation axis.

self-verification (Fu et al., 2025) and weak non-synthetic data (Amin et al., 2025). Distribution divergence measures (KL divergence, maximum mean discrepancy) offer another detection path but require a reference distribution and do not reveal the geometric structure of collapse. We complement these output-based diagnoses by tracking collapse through the spectral geometry of internal representations.

Spectral Analysis of Representations. Random matrix theory reveals implicit regularization in neural network eigenspectra (Martin and Mahoney, 2021); effective rank (Roy and Vetterli, 2007) and RankMe (Garrido et al., 2023) quantify representational dimensionality. He et al. (2025) linked spectral collapse to plasticity loss in continual learning, and Nait Saada et al. (2025) identified rank collapse in attention matrices. Most directly relevant, Gu et al. (2026) proposed SIGMA, tracking singular-value distributions during recursive training on OPT-125M over three generations. Our SLV equals half of SIGMA's $\log\det G$; we extend SIGMA in *scale* (Pythia-410M/1B, 3–8× larger), *temporal depth* (10 vs. 3 generations), and *methodology* (SLV-based mixing control). Table 1 summarizes the positioning.

Early Warning Signals and Critical Transitions. EWS for critical transitions originate in ecology, where rising autocorrelation and variance precede regime shifts (Scheffer et al., 2009, 2012), with methods from statistical tests (Dakos et al., 2012)

	SIGMA	He+ '25	This work
Setting	Recursive	Continual	Recursive
Scale	125M	Various	410M + 1B (pilot)
Generations	3	—	10
Spectral metric	$\log\det G$	spec. norm	SLV
Phase typology	—	—	✓
Mixing / r_{crit}	—	—	✓
Controller	—	—	✓

Table 1: Positioning relative to recent spectral-collapse studies. SLV $\equiv \frac{1}{2}\log\det G$. SIGMA: Gu et al. (2026); He+: He et al. (2025).

to deep learning (Bury et al., 2021). A central tension is that stabilization interventions suppress the precursors they rely on (Scheffer et al., 2009); we encounter a stronger form where real-data mixing reverses the spectral trajectory entirely (§3.2).

Data Mixing for LLM Training. Data composition significantly affects LM quality (Longpre et al., 2024; Muennighoff et al., 2023), and several methods optimize domain mixtures (Xie et al., 2023; Ye et al., 2025; Chen et al., 2025), but evaluate mixing through loss or benchmarks. Our work reveals that real-vs-synthetic mixing reshapes the spectral geometry of representations, producing a trajectory map that loss-based metrics do not capture.

3 Method

3.1 Spectral Metrics

Given a language model at generation t, we extract last-layer embeddings for n=1,000 held-out samples to form $\mathbf{E} \in \mathbb{R}^{n\times d}$. Let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(n,d)}$ denote the singular values of $\mathbf{E}$. We track three complementary spectral indicators:

Spectral Log-Volume (SLV). SLV $= \sum_{i:\,\sigma_i>\epsilon} \log\sigma_i$, where $\epsilon = 10^{-10}$ excludes numerically zero components. When $n < d$ (as in our setting with n=1,000, d=1,024), $\mathbf{E}$ has at most n non-zero singular values; SLV thus measures the log-volume of the occupied subspace rather than the full d-dimensional space. Because it sums over all non-negligible singular values, SLV is sensitive to contraction along any spectral direction, not only the dominant ones.

Finite-Sample Considerations. Our aspect ratio $\gamma = n/d = 1000/1024 \approx 0.98$ falls in the regime where Marchenko–Pastur (MP) theory (Bai and Silverstein, 2010) predicts systematic bias in spectral statistics, but since γ is constant across all conditions, this bias cancels in cross-generation

Gen	r_{eff}	SLV	PPL	HeSw	D-4	σ_1
0	417.55	2366.68	9.55	.413	.738	1942.8
2	**496.17**	**2467.98**	13.56	.380	.505	718.7
5	479.77	2365.99	24.50	.341	.412	749.2
7	464.10	2293.24	37.82	.319	.375	788.4
10	438.05	2181.35	70.99	.296	.324	874.6

Table 2: Spectral and downstream metrics at selected generations (Pythia-410M, mean of three seeds 42, 43, 44; sample s.d. at Gen 10: r_{eff} $\pm$4.8, SLV $\pm$28.3, PPL $\pm$5.6, HeSw $\pm$.006, D-4 $\pm$.15). PPL = perplexity, HeSw = HellaSwag accuracy (0-shot), D-4 = distinct-4. Bold marks the spectral peak.

comparisons. Empirically, the three-phase trajectory is stable across $n \in \{250, 5000\}$ (SLV CV $< 0.4\%$; see Limitations); the smallest observed singular value (0.016, GPT-2-medium Gen 10) remains eight orders of magnitude above ϵ. Note that the r_{crit} definition (§3.2) depends on absolute SLV non-decrease rather than cross-condition comparison, so MP bias does not cancel by construction; however, at $\gamma \approx 0.98$ the bias magnitude is small relative to the SLV changes driving the boundary ($\sim$100 units vs. a total SLV range of $\sim$2400), and the empirical stability of trajectory shapes across sample sizes confirms that this bias does not meaningfully shift r_{crit}.

Effective Rank. $r_{\text{eff}} = \exp(H(\mathbf{p}))$, where $p_i = \sigma_i / \sum_j \sigma_j$ and H is the Shannon entropy, measuring how many dimensions carry substantial variance (uniform $\to \min(n, d)$, concentrated $\to 1$).

Leading Singular Value. σ_1 tracks the dominant spectral direction; rising σ_1 under falling r_{eff} signals representational collapse into a low-dimensional subspace.

3.2 Adaptive Mixing Controller

To test whether spectral rate-of-change signals can drive automated collapse prevention, we design a closed-loop controller that adjusts the real-data mixing ratio r_t at each generation. The update rule is:

$$r_{t+1} = \text{clip}\big(\tfrac{B_{\text{rem}}}{T-t} + k \cdot \max(0,\, \Delta_t - \tau),\; r_{\min},\; r_{\max}\big) \quad (1)$$

where B_{rem} is the remaining real-data budget (fraction of total budget not yet consumed), T is the total number of generations, τ=0.002 is a sensitivity threshold, and k=10 is the controller gain. The signal Δ_t is the baseline-normalized SLV drop: $\Delta_t = (\widetilde{S}_{t-1} - \widetilde{S}_t)/|\widetilde{S}_0|$, where $\widetilde{S}_t$ is the exponential moving average (EMA)-smoothed SLV (α=0.5) and $\widetilde{S}_0$ is initialized from the pretrained model (Gen 0, SLV=2366.68). The signed form ensures that spectral decompression ($\Delta_t < 0$) does not trigger the controller; only contraction activates the $\max(0, \cdot)$ gate in Eq. 1. The output is clipped to $[r_{\min}, r_{\max}] = [0.20, 0.80]$ and subject to $\sum_t r_t \le B_{\text{total}}$. The lower bound ensures a minimum synthetic fraction; $r_{\max}$ defines the controller's operational range.

The first term distributes the remaining budget uniformly; the second injects additional real data proportional to spectral contraction severity. We set τ=0.002 as the smallest threshold avoiding false activation during decompression, and k=10 so that a moderate signal adds approximately 2 percentage points of additional real data.

Controller Silence Condition. Define r_{crit} as the minimum constant mixing ratio such that, within T=10 generations, the EMA-smoothed SLV is non-decreasing from the second generation onward: $\widetilde{S}_t \ge \widetilde{S}_{t-1}$ for all $t \in \{2, \ldots, T\}$ (excluding the initial fine-tuning transient at t=1). For conditions with multiple seeds, the condition is evaluated on the cross-seed mean trajectory. When a constant ratio $r \ge r_{\text{crit}}$ is applied, $\Delta_t \le 0 < \tau$ for all $t \ge 2$ and any $\tau > 0$, so the controller boost is identically zero. This τ-independence means the signal attenuation cannot be resolved by lowering the sensitivity threshold; the same argument extends to any rate-of-change-based controller variant (level-based triggers and cross-layer approaches are not covered by this argument). Our experiments (§4.3) empirically bound $0.25 < r_{\text{crit}} \le 0.3$ over Gen 2–10 for Pythia-410M on OpenWebText.

Second-Order Extension. A natural extension monitors the rate of change of the controller signal itself:

$$\Delta_t^2 = \widetilde{\Delta}_{t-1} - \widetilde{\Delta}_t \quad (2)$$

where $\widetilde{\Delta}_t$ is the EMA-smoothed first-order signal (α_2=0.3). A sustained negative Δ_t^2 indicates decelerating SLV growth, a potential precursor to stagnation even when SLV is still rising. We evaluate this indicator in §4.5.

3.3 Collapse Trajectory Classification

We define the *collapse trajectory map* as the mapping from constant mixing ratio r to the SLV trajectory over T generations, classifying outcomes as *collapse* (monotonic post-decompression de-

cline), *stable* (non-decreasing), or *transitional* (non-monotonic after Gen 2). We estimate r_{crit} by identifying the lowest r for which $\widetilde{S}_t \geq \widetilde{S}_{t-1}$ holds for all $t \in \{2, \ldots, T\}$.

4 Experiments

4.1 Experimental Setup

We use the publicly released final Pythia-410M-deduped checkpoint (Biderman et al., 2023) as our base model and fine-tune on a randomly sampled 50K-subset of OpenWebText (filtered to sequences with $\geq$256 non-padding tokens). Starting from the pretrained checkpoint (generation 0), each subsequent generation fine-tunes on synthetic text produced by the previous generation, for a total of 10 recursive generations. At each generation we compute the spectral indicators from §3.1 on 1K held-out last-layer embeddings, paired with three downstream metrics: perplexity, HellaSwag accuracy (0-shot), and distinct-4. These three metrics (language modeling capability, natural language understanding, and generation diversity) cover the primary evaluation dimensions accessible to a 410M-parameter model. Training hyperparameters and compute details are in Appendix D.

For the mixing comparison (§4.3), we evaluate three strategies: (1) **pure collapse** (r_t=0), (2) **fixed mix** (r_t=0.50), and (3) **adaptive** (k=10, Eq. 1). To construct the collapse trajectory map (§4.5), we additionally evaluate fixed mixing at $r \in \{0.2, 0.25, 0.3\}$.

4.2 Spectral Collapse Characterization

Table 2 reports spectral and downstream metrics across 10 recursive generations, which reveal a three-phase structure.

Phase I: Spectral Decompression (Gen 0–2). r_{eff} rises from 417.55 to 496.2 (+18.8%) and SLV from 2366.7 to 2468.0 (+4.3%), while σ_1 drops from 1942.8 to 718.7. The peak at Gen 2 is consistent across all three seeds (Figure 3; SLV coefficient of variation (CV) $<$ 1.3% at all generations; full per-generation statistics in Appendix Table 7).[2] However, this decompression does not indicate improved model quality: HellaSwag drops 8.0% and distinct-4 falls 31.6% during the same interval (Figure 4), and SLV computation takes $<$ 1 s vs. $\sim$70 min for benchmarks.

[2]With only n=3 seeds, bootstrap CIs (B=10,000, percentile method) provide descriptive summaries of observed variability; actual coverage may fall below nominal.

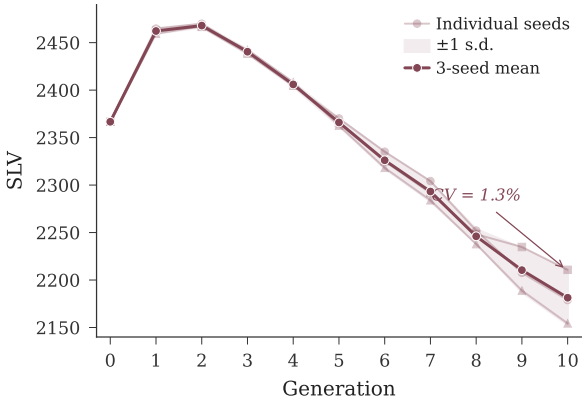


Figure 3: Cross-seed reproducibility of SLV under pure-synthetic training (r=0). Thin lines: individual seeds (42, 43, 44); thick line: 3-seed mean; shaded band: ±1 s.d. The coefficient of variation remains below 1.3% at all generations; spectral collapse follows a deterministic trajectory.

Phase II: Collapse Onset (Gen 2–4). Both r_{eff} and SLV reverse after Gen 2 (r_{eff}: 496.2→493.6, SLV: 2468→2440), marking the transition from decompression to collapse while perplexity continues its monotonic rise. SLV's post-peak decline is sharper than r_{eff}'s (−1.1% vs. −0.5% in the first post-peak generation) because SLV captures distributed contraction across all spectral directions rather than the dominant ones alone. At 1B scale, this sensitivity gap widens to a timing difference: SLV peaks one generation earlier than r_{eff} (see Cross-Scale Pilot below).

Phase III: Accelerating Collapse (Gen 4–10). SLV contraction accelerates: the per-generation decline averages −31.0 during onset (Gen 2–4) vs. −37.4 during acceleration (Gen 4–10), with 78% of the total peak-to-final decline occurring in Phase III (Table 7). By Gen 10, r_{eff} falls to 438.1 (−11.7% from peak), SLV to 2181.4 (−11.6% from peak), and perplexity reaches 71.0 (+643%). Concurrently, σ_1 traces a U-shaped trajectory: after dropping from 1942.8 to 718.7 during decompression, it rises to 874.6 by Gen 10 (Figure 1). This simultaneous σ_1 recovery and r_{eff} decline is a geometric signature of mode collapse: representations concentrate along fewer dominant directions while the overall occupied volume shrinks.

Cross-Scale Pilot. Repeating the protocol on Pythia-1B-deduped (Gen 0–3) reproduces the decompression-onset structure: SLV rises from 2835 to 2906 at Gen 1 (+2.5%) before declining to 2865 at Gen 3, while r_{eff} peaks one generation

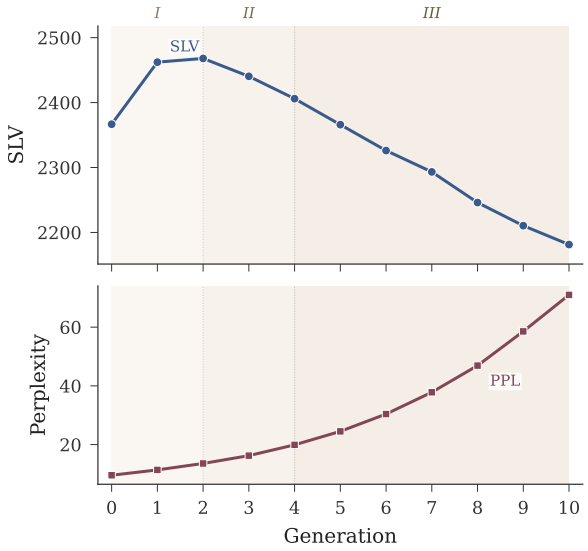


Figure 4: SLV (top) and perplexity (bottom) across 10 recursive generations (r=0, 3-seed mean). In Phase I (Gen 0–2), both metrics rise simultaneously. The SLV increase reflects distributional broadening, not model health. From Phase III onward (Gen 4–10), the two metrics diverge: SLV declines monotonically while perplexity accelerates, which confirms that post-decompression SLV faithfully tracks structural collapse.

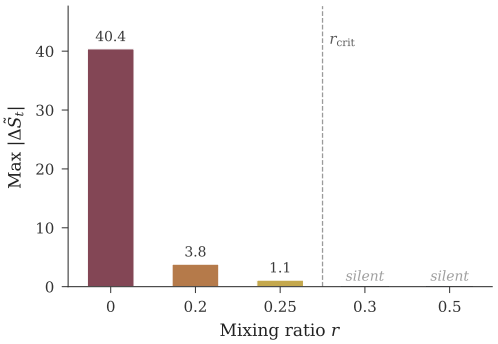


Figure 5: Maximum controller signal magnitude ($\max|\Delta\tilde{S}_t|$, $t \geq 2$) across mixing ratios. The signal drops 10× from r=0 to r=0.2 and is exactly zero for $r \geq 0.3$ ("silent"), rate-based controllers are therefore ineffective.

later at Gen 2 (580.9 vs. 570.8 at Gen 1). This one-generation SLV lead over r_{eff} (absent at 410M, where both peak simultaneously) suggests that SLV sensitivity increases with model scale. Perplexity rises monotonically throughout (8.1→13.4, +66%)—the same downstream-spectral asymmetry observed at 410M scale. Full 10-generation characterization at 1B remains future work.

4.3 Adaptive Mixing Comparison

We evaluate the controller retrospectively by computing its signal on pre-existing fixed-ratio trajectories rather than deploying it in a closed loop; the goal is to characterize signal attenuation, not to propose operational deployment. Under pure synthetic training, SLV peaks at Gen 2 (2468) and declines monotonically; with 50% mixing, SLV rises to 2572 at Gen 10, with only a negligible Gen 1 dip ($\Delta_1 \approx 0.0005$, well below τ). The adaptive controller (k=10) remains silent through all ten generations. Figure 5 quantifies the signal attenuation: the maximum controller signal magnitude drops from 40.4 at r=0 to 3.8 at r=0.2 and reaches exactly zero at $r \geq 0.3$. Even moderate mixing suffices to suppress the rate-based signal (Scheffer et al., 2009).

Despite sustained spectral expansion at r=0.5, downstream metrics degrade: perplexity increases 58% and distinct-4 falls to .152 (below even pure collapse), reinforcing that spectral health does not guarantee model health.

4.4 Alternative EWS Comparison

We compare SLV against seven alternative collapse indicators from three signal sources: embedding geometry (RankMe, isotropy), attention weights (stable rank, spectral norm), and output distributions (token entropy, raw perplexity, perplexity growth rate). We exclude generated-text metrics (distinct-n, self-BLEU) because they require full decoding (∼70 min vs. $<$1 s) and operate in output space rather than the representation space this study targets.

Table 3 summarizes the comparison under pure-synthetic training (r=0); full tabular data appear in Appendix Table 6. Only SLV and RankMe (Garrido et al., 2023) exhibit a decompression peak at Gen 2 followed by monotonic decline; all other indicators change monotonically from Gen 0 and cannot distinguish collapse onset from the initial fine-tuning transient. SLV's signal-to-noise ratio is 2.5× that of RankMe (11.8% vs. 4.7% relative change).

Seven of eight indicators reverse direction under mixing (Table 3, last column); raw perplexity is the sole exception. This near-universality confirms that signal attenuation is not specific to SLV: any rate-based controller will fall silent when mixing exceeds r_{crit}. SLV is the only indicator that combines decompression-phase sensitivity, post-peak monotonicity, and sub-second computation cost; it shares a metric foundation with SIGMA's $\log\det G$ (= 2 × SLV; §2). Raw perplexity is a

Indicator	Direction (r=0)	Rel Δ	Rev.
SLV	Peak G2, mono. ↓	11.8%	✓
RankMe	Peak G2, mono. ↓	4.7%	✓
Token ent.	Mono. ↓	9.6%	✓
Stable rank	Mono. ↓	14.0%	✓
Spec. norm	Mono. ↑	8.0%	✓
Isotropy	Non-mono. (6 rev.)	75%	✓
Raw PPL	Mono. ↑	704%	×
PPL growth	∼Mono. ↓	—	✓

Table 3: Eight EWS indicators under pure-synthetic training (r=0, Pythia-410M). **Direction**: trajectory shape (Peak Gn = decompression peak at generation n; mono. = monotonic post-peak or throughout). **Rel Δ**: total relative change. **Rev.**: whether the indicator reverses under r=0.5 mixing. Seven of eight reverse; raw PPL is the sole exception. Full data in Appendix Table 6.

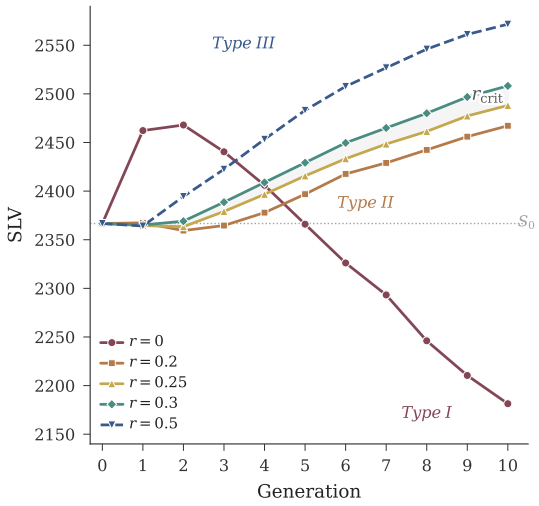


Figure 6: SLV trajectories across mixing ratios (Pythia-410M; r=0, r=0.2, r=0.25, r=0.3: 3-seed mean, solid lines; r=0.5: 2-seed mean, dashed line). Type I (r=0): decompression then monotonic collapse. Type II (r=0.2, r=0.25): transient dip followed by recovery. Type III (r=0.3, r=0.5): monotonic expansion from Gen 2 onward. The shaded band marks the critical threshold $r_{\text{crit}} \in (0.25, 0.3]$. The dashed horizontal line indicates the Gen 0 baseline.

stronger unconditional monitor (monotonic from Gen 0, no decompression mislead) but does not reveal the spectral geometry of collapse; SLV and PPL are complementary.

4.5 Collapse Trajectory Classification

We now map SLV trajectories across mixing ratios to construct the collapse trajectory map defined in §3.3. Figure 6 plots SLV across 10 recursive generations for five mixing ratios; three qualitatively distinct trajectory types emerge:

Type I: Pure Collapse (r=0). SLV exhibits large decompression (+96 points at Gen 1, peaking at Gen 2) followed by accelerating contraction (§4.2).

Type II: Near-critical Mixing (r=0.2, r=0.25). At r=0.2, SLV dips −7.3 points at Gen 2 before recovering to 2467.2 at Gen 10 (+4.3% above baseline; full per-generation data in Appendix Table 8). The EMA-smoothed SLV violates non-decreasing at Gen 2, placing r=0.2 below r_{crit}, yet the smoothed signal ($\max \Delta_t = 0.0016$, below τ=0.002) fails to trigger the controller. At r=0.25, the same pattern holds with a weaker dip (−3.3) and ∼2× weaker signal. As Figure 7 shows, r_{eff} rises monotonically for all $r \geq 0.2$, yielding only a 2-type classification (collapse vs. non-collapse); SLV's transient dip uniquely identifies the transitional regime.

Type III: Super-critical Mixing ($r \geq 0.3$). SLV rises monotonically from Gen 2 onward. At r=0.3, SLV reaches 2508.2 at Gen 10 (+6.0% above baseline); at r=0.5, 2572 (+8.7%). The critical threshold is robust to EMA smoothing ($\alpha \in \{0.3, 0.5, 0.7\}$ all yield $r_{\text{crit}} \in (0.25, 0.3]$; Appendix Table 9), and the maximum SLV departure below baseline (multi-seed mean) shrinks continuously with r (−7.3 at r=0.2, −3.3 at r=0.25, −1.5 at r=0.3), consistent with a continuous crossover. r_{crit} thus marks a transition in trajectory shape, not in final outcome: Type II trajectories recover spectrally but exhibit an initial contraction transient that Type III trajectories lack.

Output Diversity Divergence. D-4 does not increase monotonically with r: it remains highest around $r \in \{0.2, 0.25\}$ (mean .324–.342, with high cross-seed variance s.d. ≈ 0.17) before dropping to .152 at r=0.5: spectral expansion does not guarantee diversity. D-4 also shows substantially higher cross-seed variance than SLV (CV = 3.9% vs. 0.12% at r=0.25); the two metrics capture orthogonal aspects of model behavior.

Table 4 consolidates the final-generation metrics across all five mixing ratios.

Practical Implications. Neither second-order acceleration (Δ_t^2, Eq. 2) nor level-based triggers resolve the attenuation problem: Δ_t^2 is negative in late generations regardless of r, and level-based triggers fail under Type III. The trajectory map provides a direct prescription: setting $r > 0.3$ prevents spectral collapse without per-generation monitor-

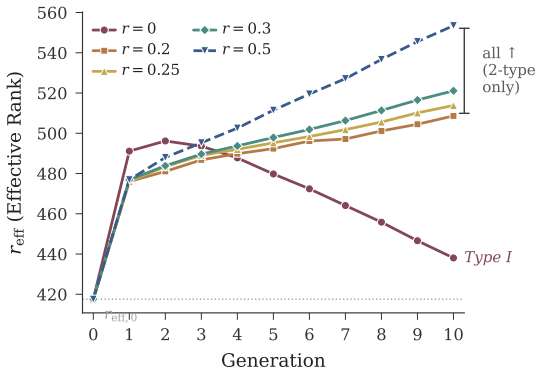


Figure 7: Effective rank (r_{eff}) trajectories across mixing ratios. Unlike SLV (Figure 6), r_{eff} rises monotonically for all $r \geq 0.2$, providing only a 2-type classification and failing to identify the Type II transitional regime.

r	Type	SLV	r_{eff}	PPL	D-4	$\max \Delta_t$	Gen.
0	I	2181.4	438.1	71.0	.324	—	10
0.20	II	2467.2	508.1	13.54	.324	.0016	10
0.25	II	2488.1	513.8	13.41	.342	.0005	10
0.30	III	2508.2	521.1	13.34	.295	0	10
0.50	III	2571.6	553.5	15.11	.152	0	10

Table 4: Final-generation spectral and downstream metrics across mixing ratios (all runs to Gen 10). PPL = perplexity, D-4 = distinct-4. $\max \Delta_t$ computed for $t \geq 2$, excluding the initial fine-tuning transient. Type classification follows §3.3. r=0 reports the mean of three seeds (42, 43, 44); r=0.3 the mean of three seeds (42, 43, 44); r=0.2 the mean of three seeds (42, 43, 44); r=0.5 the mean of two seeds (42, 43; n=2 precludes reliable uncertainty estimation); r=0.25 the mean of three seeds (42, 43, 44). D-4 exhibits high cross-seed variance at r=0 (s.d. = 0.15) and r=0.3 (s.d. = 0.155).

ing.

4.6 Cross-Architecture Validation

To test whether the three-phase trajectory generalizes beyond Pythia, we repeat the recursive pipeline on GPT-2-medium (Radford et al., 2019) (355M parameters, different pretraining corpus and tokenizer) under four conditions: pure synthetic (r=0, 9 generations), intermediate mixing (r=0.25 and r=0.3, 3 generations each), and a real-data control (r=1.0, 3 generations).

Under pure synthetic training, GPT-2-medium reproduces the three-phase pattern (Table 5): SLV peaks at Gen 2 (924, +34.1% from baseline) and declines to 415 by Gen 9 (−55.1% from peak), while perplexity rises from 15.5 to 329 (+2028%). The σ_1 trajectory follows the same U-shape (6598→1191 during decompression, rising to ∼1570 by Gen 9); dominant-direction reconcentration under falling effective rank is architecture-independent. The collapse magnitude is substantially larger than Pythia's (−55.1% vs. −11.6% SLV decline), consistent with GPT-2-medium's more anisotropic initial spectrum (r_{eff}=23 vs. Pythia's 418).

r	Type	SLV_0	SLV_{end}	ΔSLV	Match
0	I	689	415 (G9)	−39.8%	✓
0.25	—	689	738 (G3)	+7.0%	†
0.30	III	689	737 (G3)	+6.9%	✓
1.0	Ctrl	689	813 (G3)	+17.9%	✓

Table 5: Cross-architecture validation on GPT-2-medium (355M, seed 42). **Match**: whether the trajectory type matches the corresponding Pythia-410M classification. † Three-generation horizon does not resolve the Type II/III distinction at this ratio.

The real-data control (r=1.0) shows monotonic expansion with perplexity *decreasing* (−27.4%), which confirms that collapse is specific to recursive synthetic training.

At intermediate ratios (r=0.25, r=0.3), both conditions reproduce the decompression phase (Gen 0→1: SLV +7.5–8.5%, r_{eff} from 23 to 136–142, σ_1 −80–82%) and stabilize through Gen 3, with EMA-smoothed SLV monotonically non-decreasing. At r=0.3 this matches Pythia's Type III classification. At r=0.25, the three-generation window does not resolve the Type II/III distinction, since Pythia's characteristic EMA dip appears only at Gen 5+; the trajectory is consistent with the Pythia-410M phase boundary but does not independently confirm r_{crit} for this architecture.

5 Discussion and Conclusion

That downstream performance degrades during spectral expansion confirms these are collapse indicators, not quality measures; SLV and perplexity are complementary. Our analysis reveals a reproducible three-phase collapse trajectory, identifies SLV as a cheap spectral indicator, and uncovers a prevention–detectability tradeoff: real-data mixing reverses the spectral trajectory and the controller signal vanishes at $r \geq 0.3$, so rate-of-change-based reactive controllers cannot function where prevention succeeds. The trajectory map localizes $r_{\text{crit}} \in (0.25, 0.3]$; setting $r > r_{\text{crit}}$ prevents collapse without per-generation monitoring.

Limitations

Scale and Scope. We validate at two model scales (Pythia-410M and a 1B pilot through Gen 3) with all spectral metrics computed from last-layer embeddings only; intermediate layers may exhibit different dynamics. The 1B pilot confirms the decompression peak but is insufficient to characterize full acceleration or estimate r_{crit} at that scale. All fine-tuning uses a single domain (OpenWebText, English web text); spectral collapse dynamics may differ for code, multilingual, or domain-specific corpora.

Temporal Horizon. Ten recursive generations may be shorter than production pipelines. Type II (transitional) trajectories may eventually revert to collapse at longer horizons, which would shift r_{crit} upward. The proportional controller uses a single gain k without integral or derivative terms; a more sophisticated design might partially mitigate signal attenuation.

Generalization. Signal attenuation may extend to implicit-loop settings where web corpora contain model-generated content, but validating this requires controlled experiments with web-scale data. The prevention–detectability tradeoff is established for rate-based spectral signals; information sources outside the embedding spectrum (e.g., generated-text analysis) may exhibit different attenuation profiles.

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A Alternative EWS Indicator Details

Table 6 provides full tabular data for the eight alternative EWS indicators compared in §4.4.

Indicator	Source	Peak	Mono.	Rel Δ	Rev.
SLV	emb-SVD	Gen 2	✓	11.8%	✓
RankMe	emb-SVD	Gen 2	✓	4.7%	✓
Token ent.	logit	—	✓	9.6%	✓
Stable rank	attn wt.	—	✓	14.0%	✓
Spec. norm	attn wt.	—	✓	8.0%	✓
Isotropy	emb	—	× (6)	75%	✓
Raw PPL	eval	—	✓	704%	×
PPL growth	ppl	—	∼(1)	—	✓

Table 6: Alternative early warning indicators under pure-synthetic training (r=0, seed 42, Pythia-410M, 10 gen). **Peak**: generation at which the indicator reverses (— = monotonic from Gen 0). **Mono.**: post-peak monotonicity (✓ = yes; number = violations). **Rel** Δ: magnitude of total relative change; for peaked indicators (SLV, RankMe), computed as $|v_{10} - v_{\text{peak}}|/|v_{\text{peak}}|$; for monotonic indicators, $|v_{10} - v_0|/|v_0|$. **Rev.**: direction reversal under r=0.5 mixing (✓ = reversed). Note: Raw PPL Rel Δ is 704% for seed 42; the main text reports 643% (3-seed mean).

B Per-Generation Cross-Seed Variance

Table 7 reports all six metrics at every generation under pure-synthetic training (r=0), with mean and sample standard deviation across three seeds (42, 43, 44). Spectral metrics (r_{eff}, SLV, σ_1) exhibit low cross-seed variance throughout the trajectory (SLV coefficient of variation $< 1.3\%$ at all generations); spectral collapse follows a deterministic geometric trajectory. In contrast, D-4 variance increases sharply after Gen 5, reaching a coefficient

of variation of 46% at Gen 10, consistent with the stochastic nature of n-gram diversity under progressive mode collapse.

Table 8 provides the same breakdown under transitional mixing (r=0.2), where D-4 variance is the dominant source of cross-seed disagreement.

C Controller Signal Sensitivity to EMA Smoothing Parameter

α	r=0	r=0.2	r=0.25	r=0.3	r=0.5
0.3	.0141	.0009	.0003	0	0
0.5	.0171	.0016	.0005	0	0
0.7	.0183	.0023	.0005	0	0

Table 9: Normalized maximum controller signal magnitude ($\max|\Delta\tilde{S}_t|/\tilde{S}_t$, $t \geq 2$) across mixing ratios and EMA smoothing parameters. The critical threshold $r_{\text{crit}} \in (0.25, 0.3]$ is invariant to α: signal is exactly zero for $r \geq 0.3$ at all three smoothing values, the phase boundary is a structural property of the trajectory rather than an artifact of smoothing.

D Training and Compute Details

Fine-tuning uses AdamW (lr 5×10^{-5}, weight decay 0.01, linear warmup 5%, effective batch size 32) for one epoch per generation, with gradient clipping at max norm 1.0. Synthetic text is generated with temperature 1.0, top-k=50, max length 512. All experiments run on NVIDIA RTX PRO 6000 GPUs (96 GB); each 10-generation pipeline requires 15–26 single-GPU hours depending on the mixing ratio, totaling approximately 300 GPU-hours across all reported experiments.

E Use of AI Assistance

We used Claude for coding, code drafting, debugging, and language polishing.

Table 7: Per-generation metrics under pure-synthetic training (r=0, Pythia-410M, mean$_{\pm \text{s.d.}}$ across seeds 42, 43, 44). Gen 0 shares a single pretrained checkpoint; D-4 variance at Gen 0 reflects generation sampling. Bold marks the spectral peak.

Gen	r_{eff}	SLV	PPL	HeSw	D-4	σ_1
0	$417.5_{\pm 0.0}$	$2366.7_{\pm 0.0}$	$9.55_{\pm 0.00}$	$.413_{\pm .000}$	$.738_{\pm .006}$	$1942.8_{\pm 0.0}$
1	$491.1_{\pm 0.6}$	$2462.3_{\pm 2.7}$	$11.37_{\pm 0.01}$	$.390_{\pm .001}$	$.609_{\pm .014}$	$808.9_{\pm 5.4}$
2	$\mathbf{496.2}_{\pm 0.3}$	$\mathbf{2468.0}_{\pm 1.7}$	$13.56_{\pm 0.02}$	$.380_{\pm .004}$	$.505_{\pm .011}$	$718.7_{\pm 7.0}$
3	$493.6_{\pm 0.4}$	$2440.4_{\pm 2.0}$	$16.23_{\pm 0.02}$	$.367_{\pm .005}$	$.465_{\pm .033}$	$708.7_{\pm 7.5}$
4	$487.8_{\pm 1.9}$	$2406.0_{\pm 2.0}$	$19.90_{\pm 0.09}$	$.354_{\pm .003}$	$.419_{\pm .047}$	$722.9_{\pm 6.9}$
5	$479.8_{\pm 0.5}$	$2366.0_{\pm 3.8}$	$24.50_{\pm 0.17}$	$.341_{\pm .003}$	$.412_{\pm .053}$	$749.2_{\pm 7.1}$
6	$472.4_{\pm 0.6}$	$2326.1_{\pm 8.6}$	$30.40_{\pm 0.45}$	$.329_{\pm .004}$	$.402_{\pm .067}$	$764.6_{\pm 5.7}$
7	$464.1_{\pm 0.2}$	$2293.2_{\pm 10.2}$	$37.82_{\pm 0.94}$	$.319_{\pm .000}$	$.375_{\pm .082}$	$788.4_{\pm 4.0}$
8	$455.8_{\pm 1.1}$	$2246.1_{\pm 7.4}$	$46.90_{\pm 2.41}$	$.311_{\pm .005}$	$.337_{\pm .137}$	$806.4_{\pm 8.0}$
9	$446.6_{\pm 2.1}$	$2210.4_{\pm 23.1}$	$58.56_{\pm 4.52}$	$.301_{\pm .003}$	$.316_{\pm .148}$	$841.1_{\pm 23.6}$
10	$438.1_{\pm 4.8}$	$2181.4_{\pm 28.3}$	$70.99_{\pm 5.56}$	$.296_{\pm .006}$	$.324_{\pm .150}$	$874.6_{\pm 44.2}$

Table 8: Per-generation metrics under transitional mixing (r=0.2, Pythia-410M, mean$_{\pm \text{s.d.}}$ across seeds 42, 43, 44). Gen 0 shares a single pretrained checkpoint; D-4 variance at Gen 0 reflects generation sampling. HellaSwag was not evaluated for mixing experiments.

Gen	r_{eff}	SLV	PPL	D-4	σ_1
0	$417.5_{\pm 0.0}$	$2366.7_{\pm 0.0}$	$9.55_{\pm 0.00}$	$.738_{\pm .006}$	$1942.8_{\pm 0.0}$
1	$476.2_{\pm 0.8}$	$2367.4_{\pm 2.1}$	$10.28_{\pm 0.00}$	$.671_{\pm .015}$	$961.8_{\pm 1.1}$
2	$481.7_{\pm 1.4}$	$2359.4_{\pm 1.5}$	$10.74_{\pm 0.01}$	$.519_{\pm .037}$	$879.2_{\pm 10.3}$
3	$486.7_{\pm 0.3}$	$2364.6_{\pm 1.8}$	$11.14_{\pm 0.01}$	$.471_{\pm .006}$	$811.9_{\pm 9.6}$
4	$489.6_{\pm 0.6}$	$2377.9_{\pm 1.2}$	$11.49_{\pm 0.00}$	$.429_{\pm .026}$	$779.2_{\pm 9.2}$
5	$492.6_{\pm 0.5}$	$2396.8_{\pm 1.8}$	$11.83_{\pm 0.00}$	$.420_{\pm .037}$	$748.8_{\pm 9.3}$
6	$495.5_{\pm 1.2}$	$2417.7_{\pm 10.5}$	$12.17_{\pm 0.02}$	$.413_{\pm .070}$	$749.7_{\pm 22.5}$
7	$496.7_{\pm 1.2}$	$2429.0_{\pm 6.4}$	$12.49_{\pm 0.05}$	$.379_{\pm .096}$	$750.9_{\pm 30.6}$
8	$499.8_{\pm 2.3}$	$2442.4_{\pm 10.3}$	$12.82_{\pm 0.12}$	$.365_{\pm .119}$	$753.9_{\pm 23.4}$
9	$503.8_{\pm 1.2}$	$2456.0_{\pm 6.3}$	$13.18_{\pm 0.18}$	$.349_{\pm .143}$	$738.1_{\pm 37.6}$
10	$508.1_{\pm 2.8}$	$2467.2_{\pm 12.0}$	$13.54_{\pm 0.25}$	$.324_{\pm .171}$	$722.7_{\pm 26.6}$